Macroeconomics I — MPE Insper

Practice list: Consumption, Saving and Labor Markets — graduate level (Listas 3–4)

Instructions. Multiple choice: mark the single correct option (no penalty for wrong answers). *True/False*: every 2 wrong items cancel 1 right one (score = $\max\{0, \text{right} - \text{wrong}/2\}\times$ item value); blank items do not count. Discursive questions: show every step. Numbers are illustrative and chosen to give clean answers.

Block 1 — Consumption and saving

(Kurlat ch. 6; Romer ch. 8 • 25 points)

Question 1. (10 points) Objective items.

a) (3 points) A household lives two periods with $u(c_1) + \beta u(c_2)$, $u = c^{1-\sigma}/(1-\sigma)$, $\sigma = 2$, $\beta = 0.95$, and income (y_1, y_2). It earns $r_s = 2\%$ (real) on deposits but pays $r_b = 30\%$ (real) on consumer credit. For which values of y_2/y_1 does it neither borrow nor lend?

A. $[0.984,\ 1.111]$

B. $[0.969,\ 1.235]$

C. $[1.020,\ 1.300]$

D. $[0.950,\ 1.000]$

b) (3 points) Three periods, $r = 0$, wealth $W = 100$, log utility. The self at date t maximizes $\ln c_t + \beta \sum_{s>t} \ln c_s$ with $\beta = 1/2$ (present bias, no long-run discounting). Self 0 anticipates how self 1 will behave. What share of consumption *in every period* would self 0 give up to commit future selves to its own preferred plan?

A. 0%

B. 2.9%

C. 5.9%

D. 1.5%

c) (2 points, T or F) In (b), a sophisticated self 0 consumes less at date 0 than a naive self 0, who believes future selves will follow its plan, because it anticipates their overconsumption.

d) (2 points, T or F) A household with $y_1 = 0$ and $y_2 > 0$ that can borrow at r lowers c_1 when r rises, whatever $\sigma > 0$.

Question 2. (15 points) Discursive. Permanent income over an infinite horizon. A household maximizes $\sum_{t\geq 0} \beta^t u(c_t)$, $u = c^{1-\sigma}/(1-\sigma)$, subject to $c_t + a_{t+1} = y_t + (1+r)a_t$, a_0 given, and the no-Ponzi condition $\lim_{T\to\infty} a_{T+1}/(1+r)^T \geq 0$. Let $R \equiv 1 + r$ and $H_t \equiv \sum_{j\geq 0} y_{t+j} R^{-j}$.

a) (3 points) Derive the intertemporal budget constraint and the consumption function $c_0 = (1-\gamma)(Ra_0 + H_0)$, $\gamma \equiv \beta^{1/\sigma} R^{1/\sigma - 1}$. State the condition for a solution and show that with log utility the propensity to consume out of wealth does not depend on r.

Solution:

b) (4 points) Let $\beta R = 1$. Show that $c_t = \frac{r}{R}(Ra_t + H_t)$ and prove Campbell's "saving for a rainy day" equation, $s_t \equiv ra_t + y_t - c_t = -\sum_{j\geq 1} R^{-j}\Delta y_{t+j}$.

Solution:

c) (4 points) Brazil's 2019 pension reform, stylized. $r = 4\%$, $\beta R = 1$, $a_0 = 0$; earnings are 100 for $t = 0, \ldots, 29$ and the pension is 60 forever from $t = 30$. At $t = 0$ a reform is announced that cuts the pension to 45. Compute Δc_0 with the consumption function and Δs_0 with Campbell's equation. Compare with the response to a one-off transfer and to a permanent raise of the same size, and give the limit of Δc_0 as $r \to 0$ (keeping $\beta R = 1$).

Solution:

d) (4 points) Now $a_{t+1} \geq 0$ for all t (no borrowing), log utility, $r = 4\%$, $\beta R = 1$, $a_0 = 0$; earnings are 50 for $t \leq 9$ and 100 from $t = 10$. Show the unconstrained plan is infeasible. Prove, with the Kuhn–Tucker conditions, that $c_t = y_t$ is optimal and find which multipliers are positive. Compute the welfare cost of the constraint (the share of the unconstrained consumption path the household would accept to give up) and the propensity to consume a small transfer received at $t = 0$ and at $t = 9$.

Solution:

Block 2 — Labor supply and search

(Kurlat ch. 7; Romer ch. 10 • 25 points)

Question 3. (10 points) Objective items.

a) (3 points) Let $u = \ln c - \psi \frac{h^{1+1/\varepsilon}}{1+1/\varepsilon}$ with $c = wh + \pi$, evaluated at $\pi = 0$, $\varepsilon = 1/2$. The Marshallian, Hicksian (compensated) and Frisch elasticities of hours with respect to w are:

A. $(0,\ 1/3,\ 1/2)$

B. $(1/2,\ 1/2,\ 1/2)$

C. $(0,\ 1/2,\ 1/3)$

D. $(-1/3,\ 0,\ 1/2)$

b) (3 points) Prescott's calculation with $u = \ln c + \alpha \ln(1-h)$, $Y = K^{\vartheta}(hN)^{1-\vartheta}$, $\vartheta = 1/3$, a labor tax wedge τ rebated lump-sum. Calibrate α on the US ($h = 0.40$, $\tau = 0.40$, $c/y = 0.75$) and predict Brazil's hours ($\tau = 0.50$, $c/y = 0.80$):

A. 0.400

B. 0.357

C. 0.342

D. 0.333

c) (2 points, T or F) With matching function $m = \mu U^{1-\alpha} V^{\alpha}$, the search equilibrium with bargained wages is efficient if and only if the workers' bargaining power equals α.

d) (2 points, T or F) In the steady-state search model with $w = (1-\phi)z + \phi(y + \kappa\theta)$, if unemployment income z and the vacancy cost κ are proportional to productivity y, a permanent rise in y leaves unemployment unchanged.

Question 4. (15 points) Discursive. The search-and-matching model in steady state (monthly). Labor force 1; unemployment u, vacancies v, tightness $\theta = v/u$; matches $m = \mu u^{1-\alpha} v^{\alpha}$; job-finding rate $f(\theta) = m/u = \mu\theta^{\alpha}$; vacancy-filling rate $q(\theta) = m/v = \mu\theta^{\alpha-1}$; separations at rate s. A filled job produces y and pays w; a vacancy costs κ per month; entry is free. Take the wage rule $w = (1-\phi)z + \phi(y + \kappa\theta)$ as given (it comes from Nash bargaining, ϕ = workers' power). Use one fact: a flow x that lasts until a Poisson event with rate ϱ is worth $x/(r+\varrho)$. Numbers: $r = 0.004$, $s = 0.03$, $\mu = 0.3$, $\alpha = 0.5$, $\phi = 0.5$, $y = 1$, $z = 0.7$.

a) (3 points) Derive the Beveridge curve from flow balance and prove that, in the (u, v) plane, it is downward sloping and convex.

Solution:

b) (4 points) Derive the job-creation condition from free entry, combine it with the wage rule, and prove that tightness θ^* exists and is unique. Calibrate κ so that $\theta^* = 1$ and compute u^* and v^*. Sign $d\theta^*/dy$, $d\theta^*/dz$, $d\theta^*/d\phi$ and $d\theta^*/d\mu$.

Solution:

c) (4 points) Efficiency with $r \to 0$. A planner chooses θ to maximize steady-state net output $y(1-u) + zu - \kappa v$ subject to the Beveridge curve. Derive its condition and prove that the market θ coincides with it if and only if $\phi = 1 - \alpha$. With κ from (b), compute efficient θ and u, and the market values when $\phi = 0.7$. Explain the externalities.

Solution:

d) (4 points) With $\alpha = 1/2$ the job-creation condition is a quadratic in $x = \sqrt{\theta}$: solve it. Compute θ^* and u^* after (i) mismatch lowers μ to 0.225 and (ii) more generous unemployment insurance raises z to 0.8. Derive the elasticity of θ^* with respect to y at the calibrated point, and of u^*. What does the formula say about Shimer's puzzle?

Solution: